Image Enhancement in Frequency Domain

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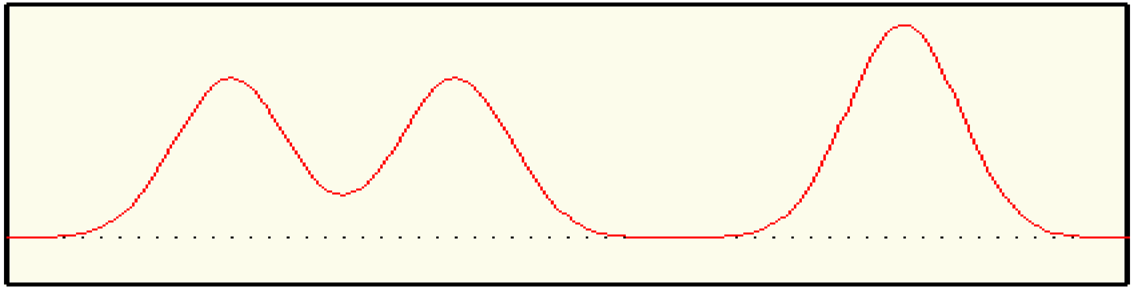
What is a Fourier transform?

Main Idea

Any function can be described by a <u>summation</u> of waves

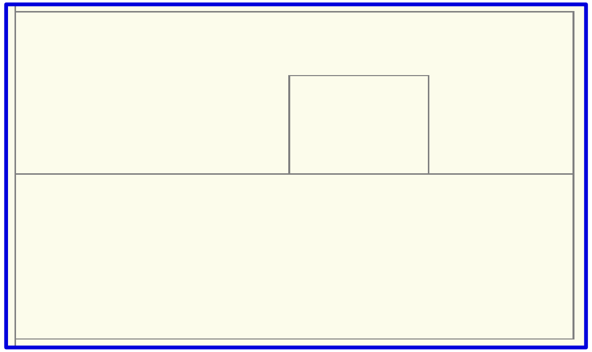
with different <u>amplitudes</u> and <u>phases</u>



Frequency Spectra

First base function

second base function



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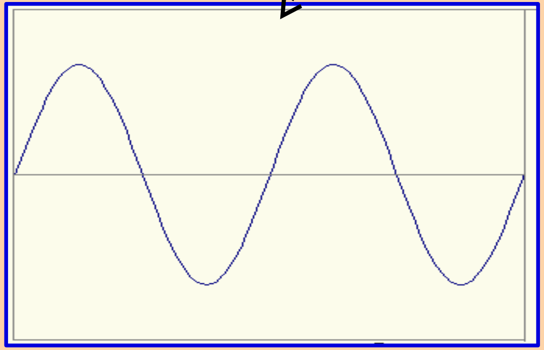

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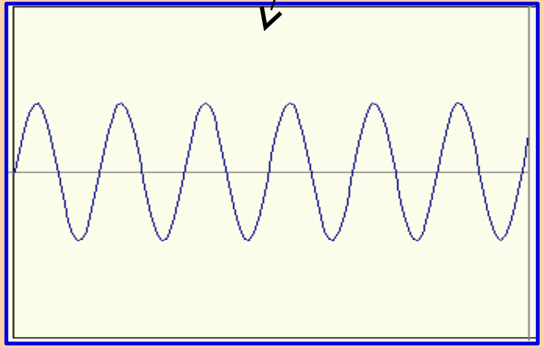

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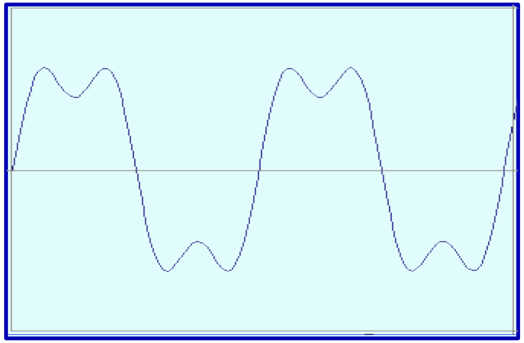
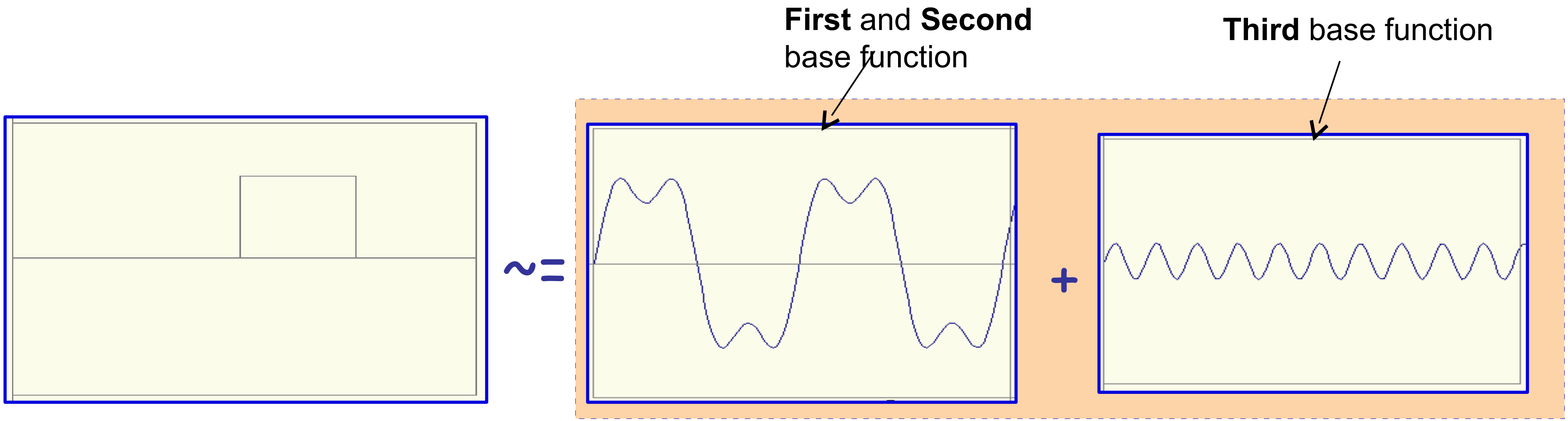
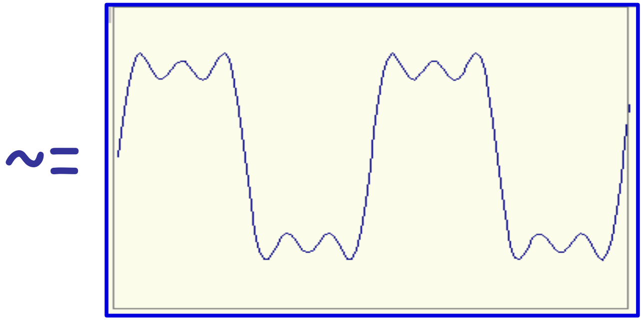


The **sum** of first and second base functions is already not bad approximation

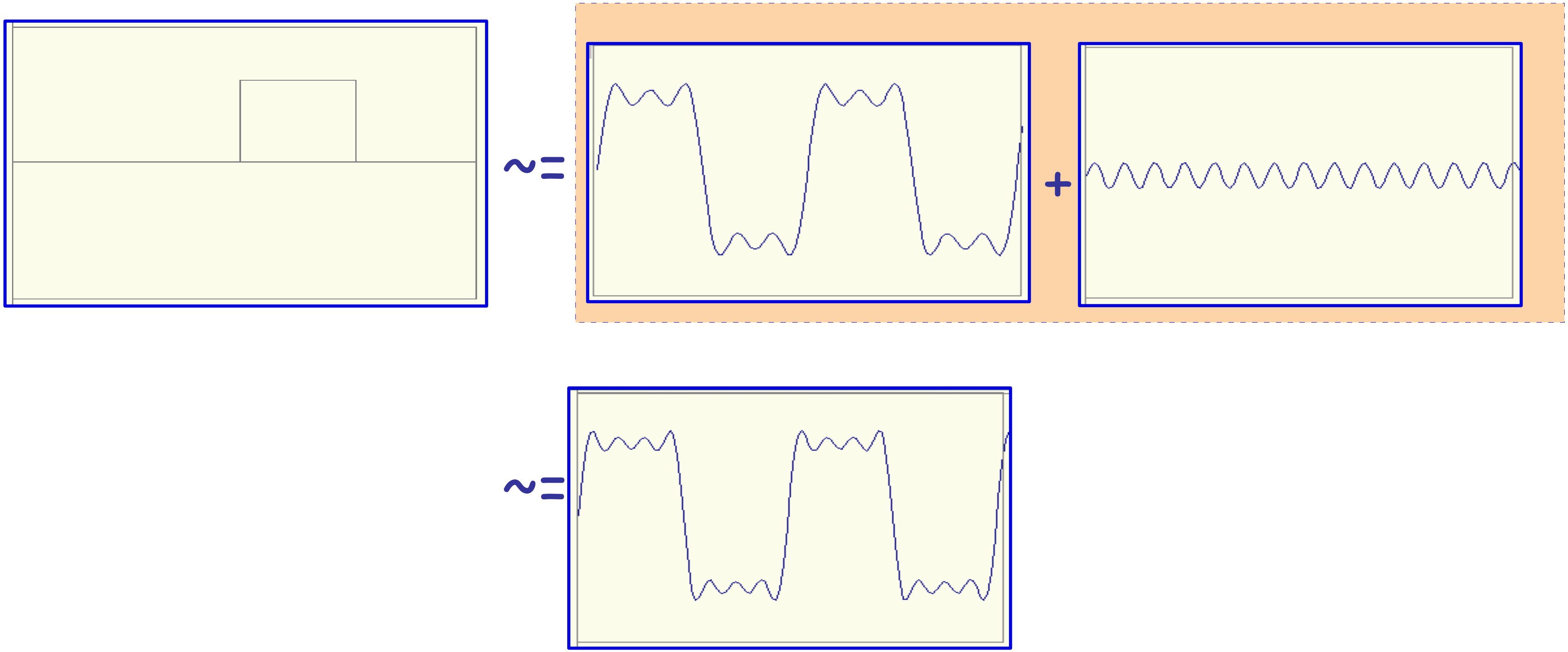
Frequency Spectra



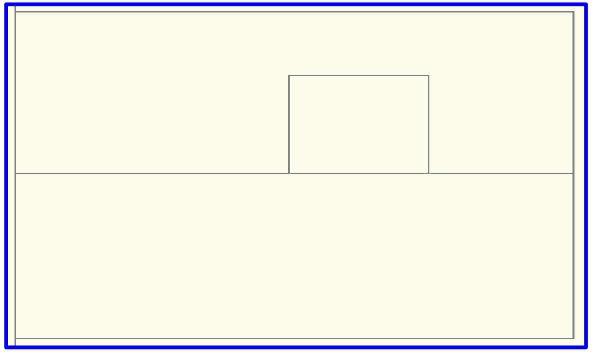


Sum of **first** , **second** and **third** base functions is even **better approximation**

Frequency Spectra



Frequency Spectra



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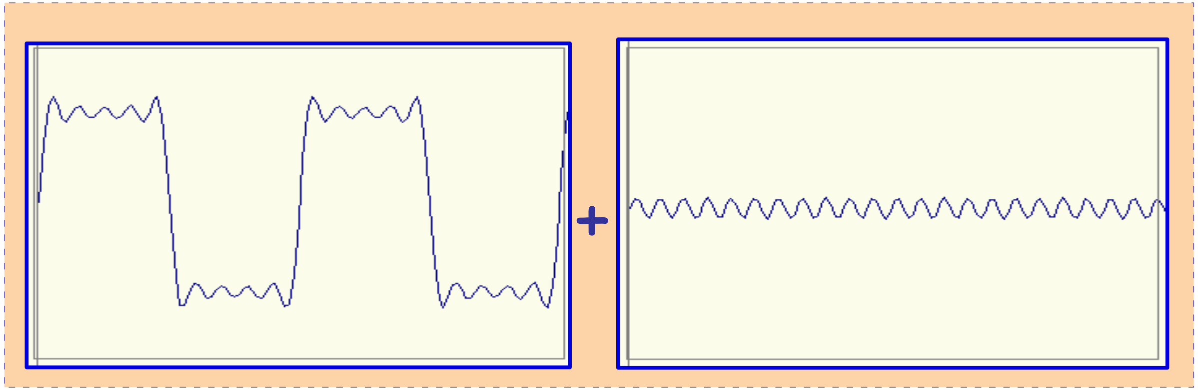

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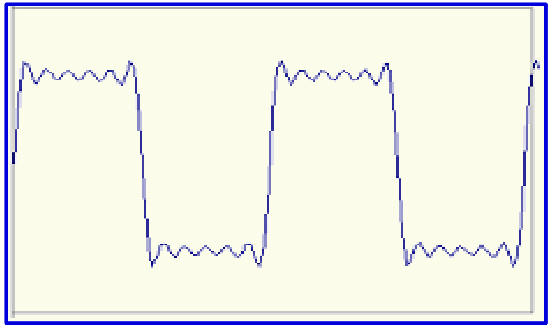
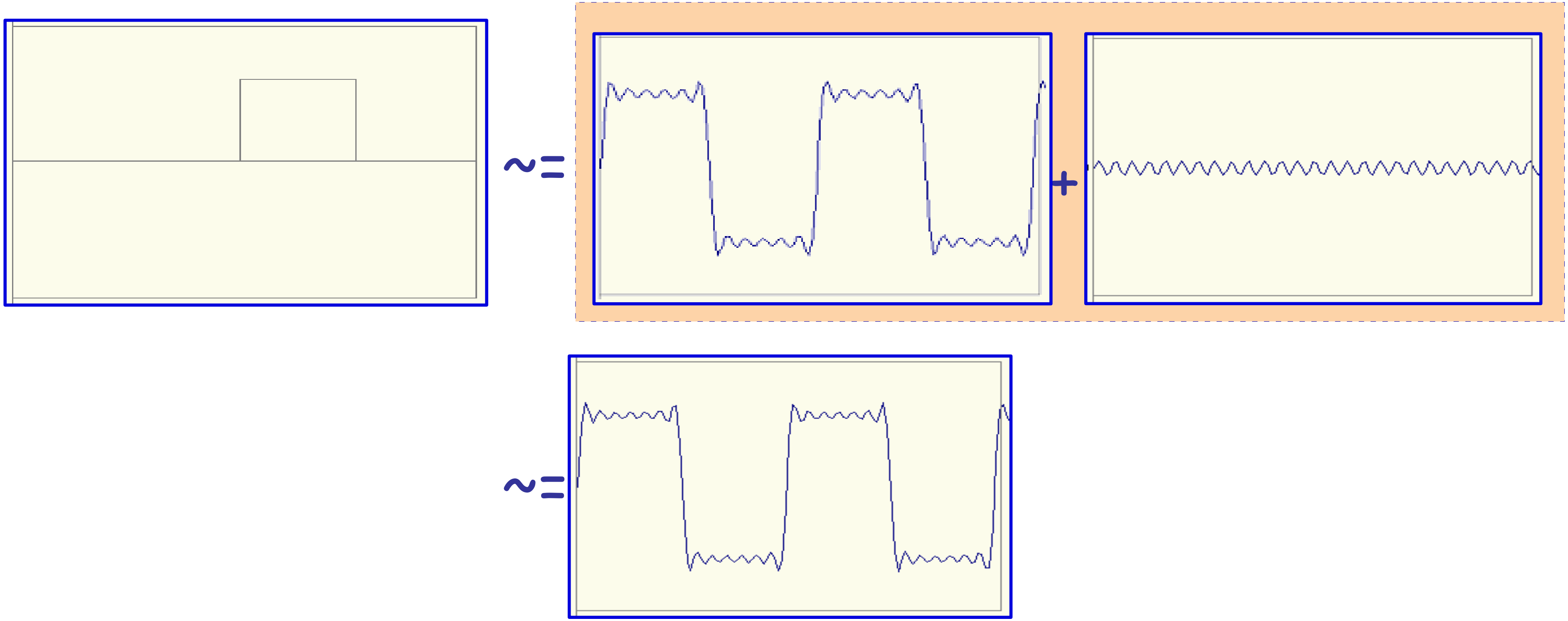


Frequency Spectra



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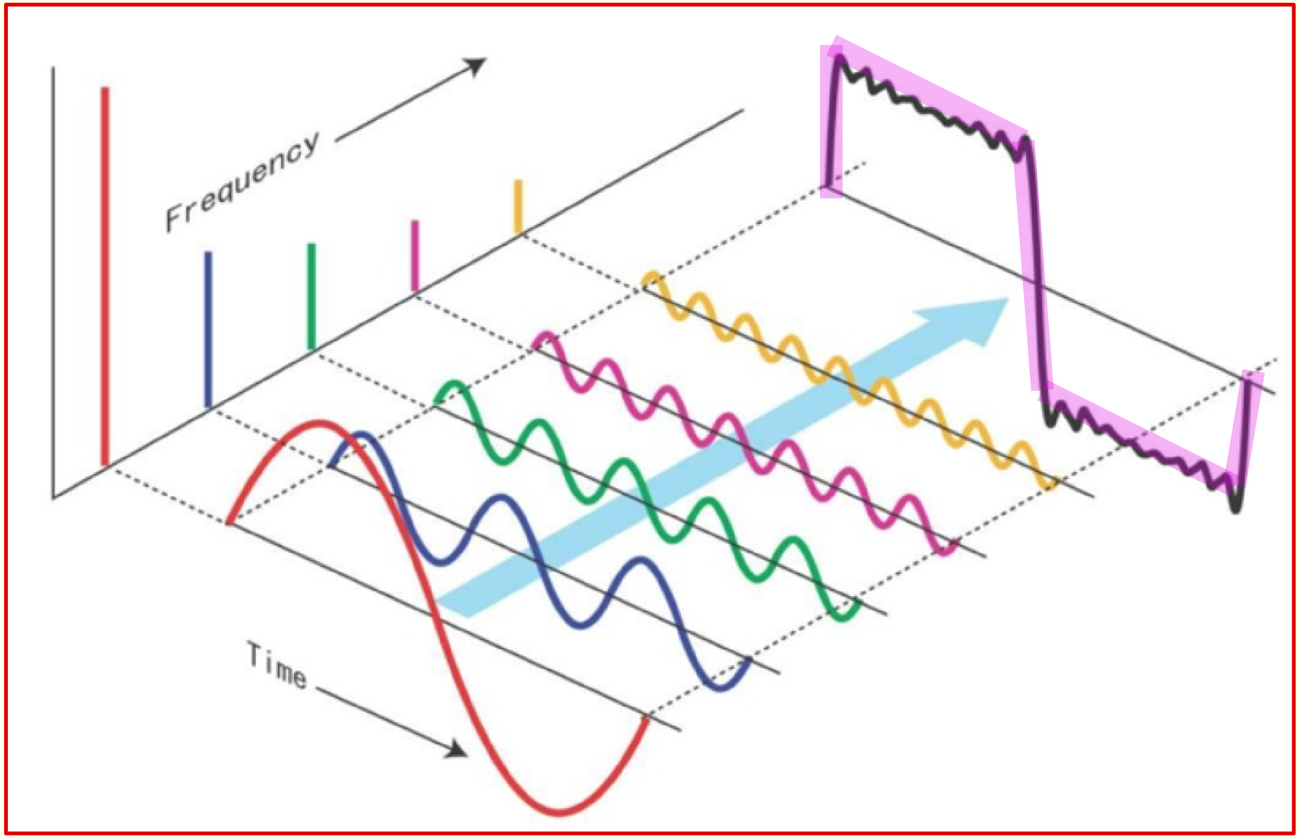
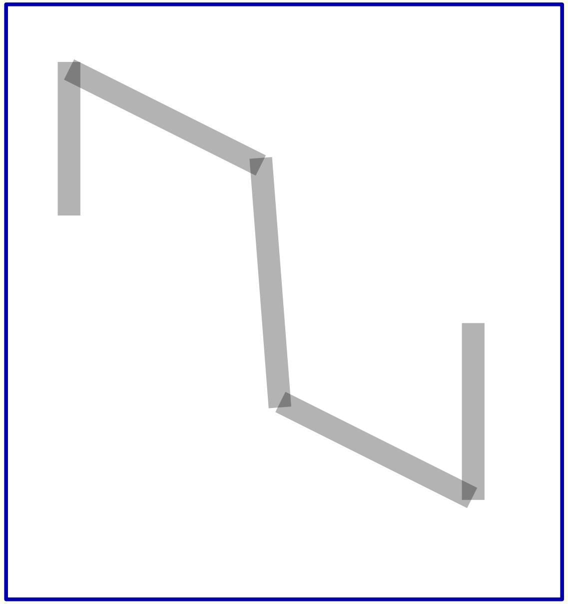
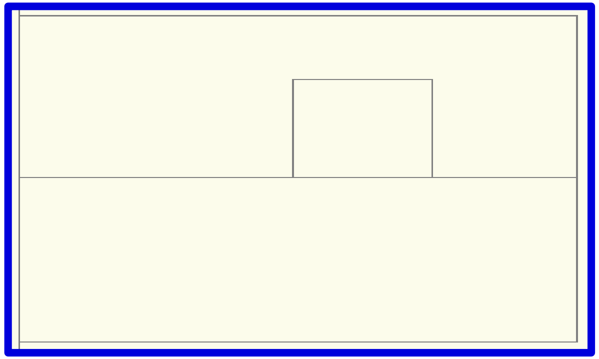
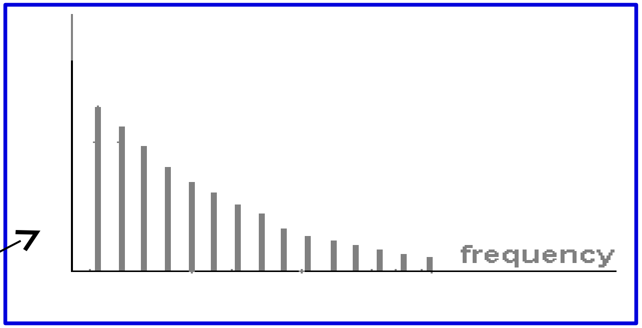




Frequency Spectra



$$= \quad A\sum_{k=1}^{\infty}\frac{1}{k}\sin(2\pi kt)$$

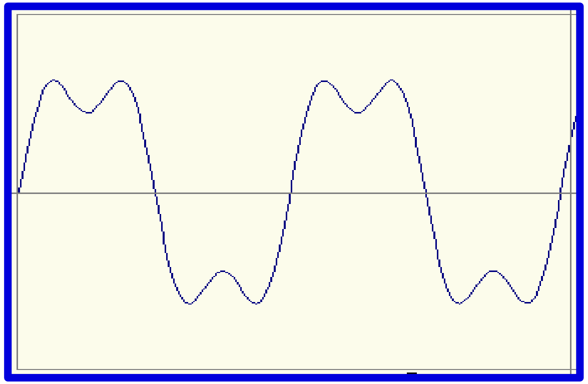


In time domain

In frequency domain

Time and Frequency

example : $g(t) = \sin(2\pi f t) + (1/3)\sin(2\pi (3f) t)$



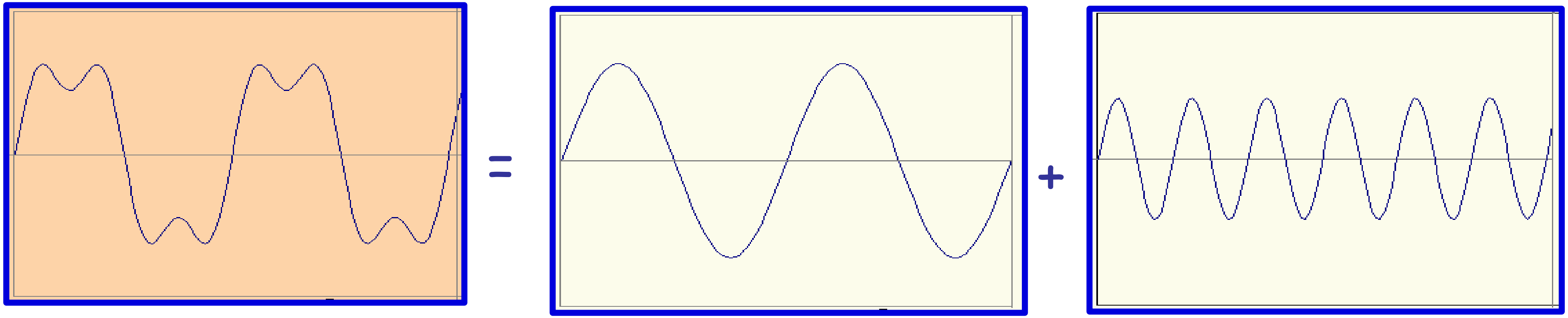
Our periodic signal is a mixture of **two frequencies (f and 3f)**

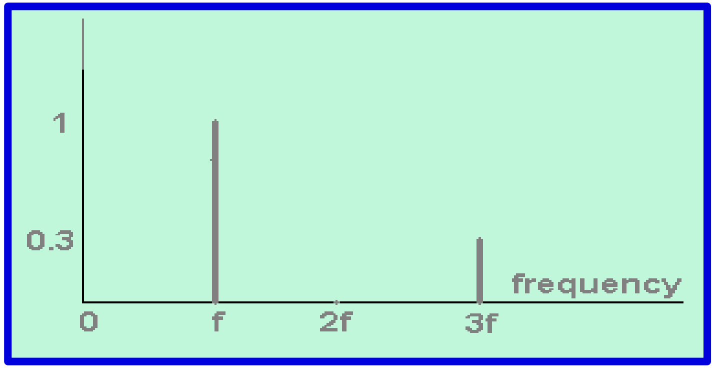
Before we go to using complex numbers and formulas let us get intuition about time and frequency

Time and Frequency

example : $g(t) = \sin(2\pi f\, t) + (1/3)\sin(2\pi\, (3f)\, t)$

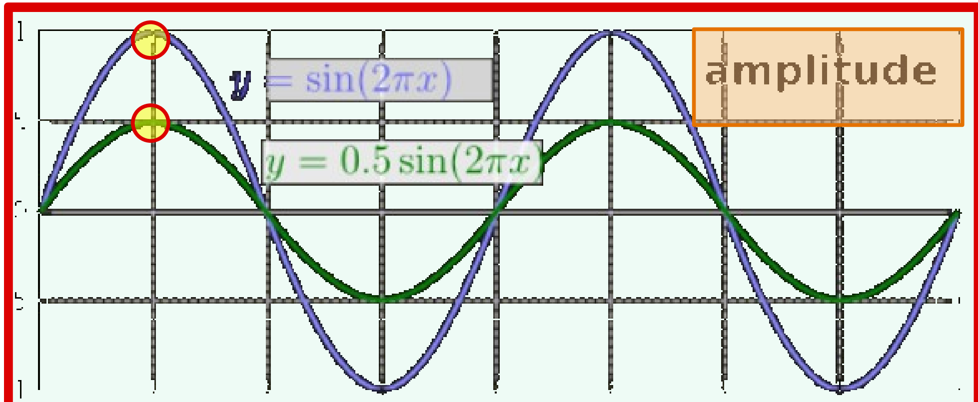
In time domain



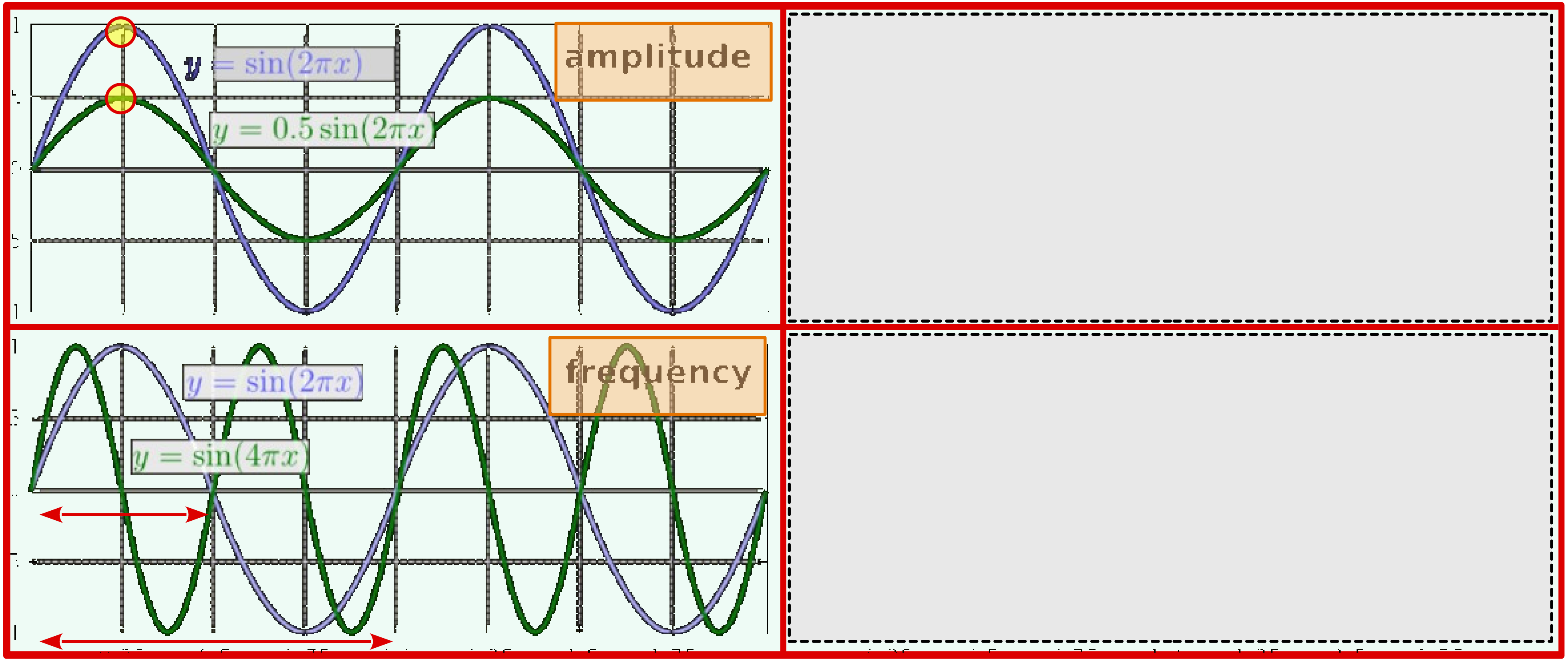


In frequency domain

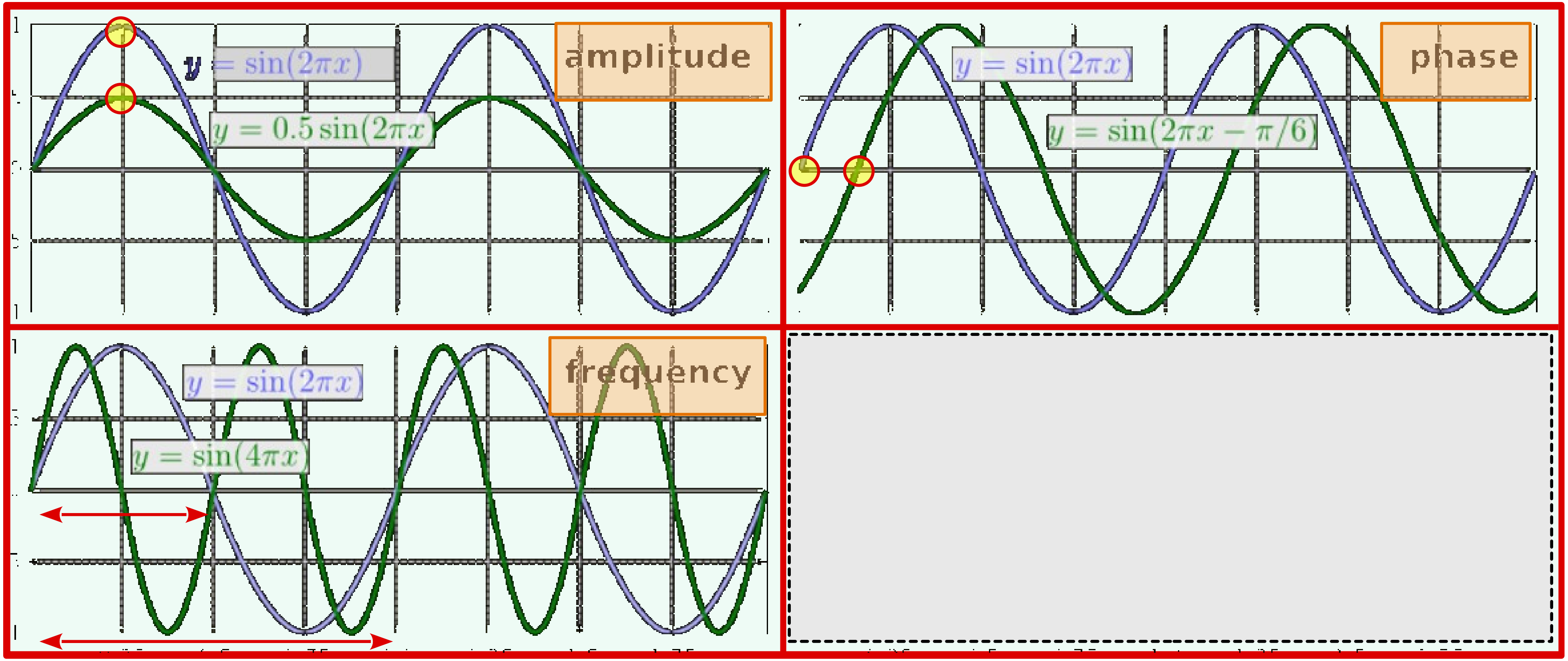
Frequency, Amplitude, and Phase



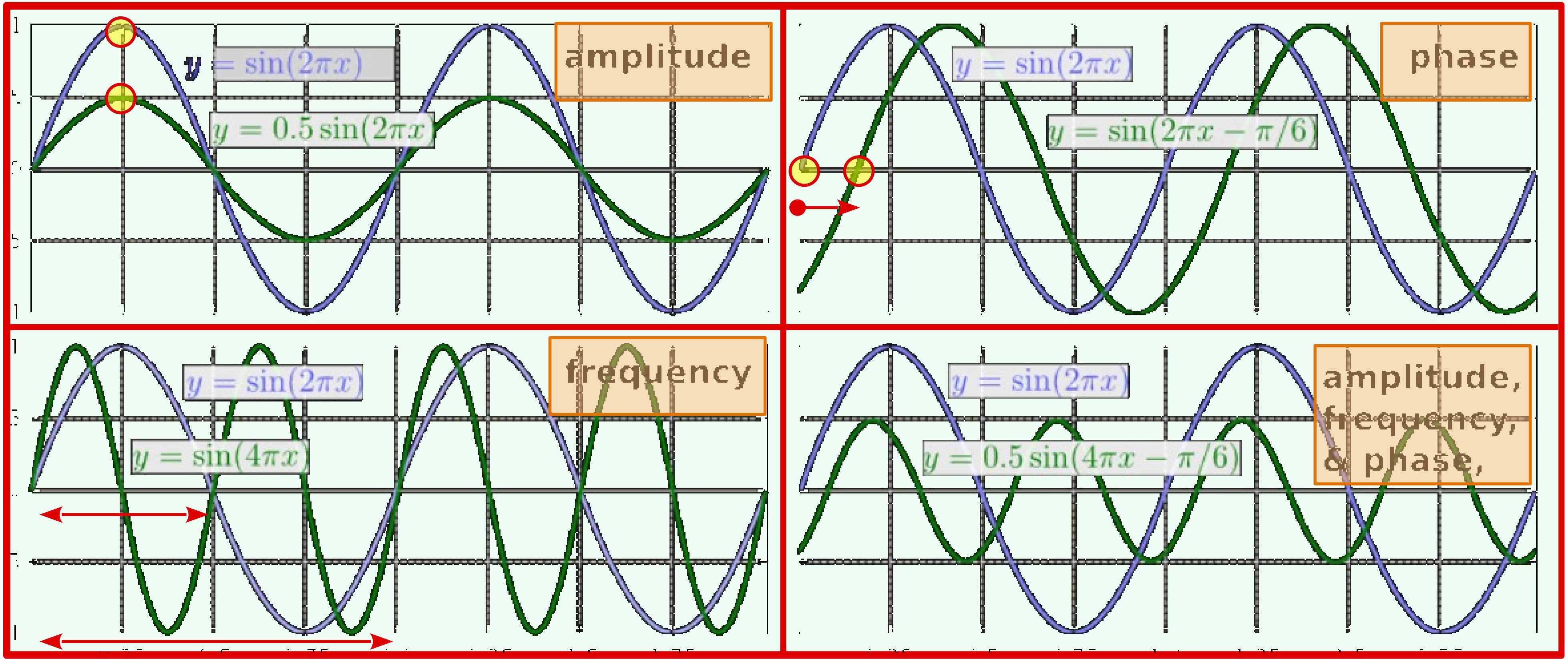
Frequency, Amplitude, and Phase



Frequency, Amplitude, and Phase

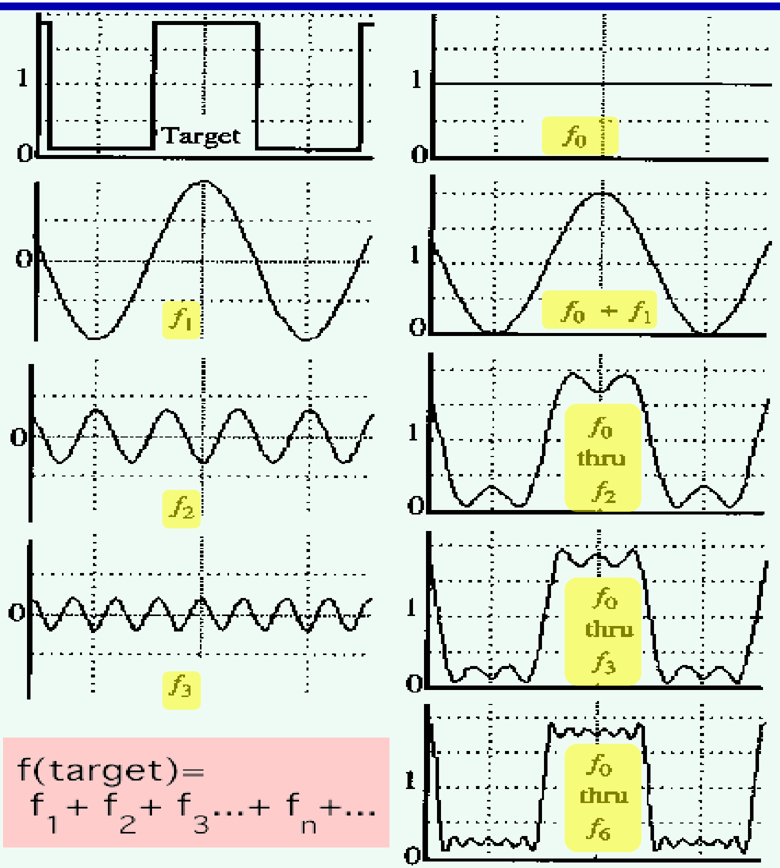


Frequency, Amplitude, and Phase



A Sum of Sinusoids

- Our building block:
- $A\sin(\omega x + \phi)$
- Add enough of them to get any signal *f(x)* you want!
- How many degrees of freedom?
- What does each control?
- Which one encodes the coarse vs. fine structure of the signal?



f(target)= $f_1 + f_2 + f_3 ... + f_n + ...$

Math Review - Complex numbers

- Complex numbers

  $4.2 + 3.7i$

  $9.4447 - 6.7i$

  $-5.2 = (-5.2 + 0i)$

- General Form

  $\mathbf{Z = a + b}i$

  $Re(Z) = a$

  $Im(Z) = b$

Real and imaginary parts

Math Review - Complex numbers

- Complex numbers

  $4.2 + 3.7i$

  $9.4447 - 6.7i$

  $-5.2 = (-5.2 + 0i)$

- General Form

  $\mathbf{Z = a + b}\boldsymbol{i}$

  $\mathrm{Re}(Z) = a$

  $\mathrm{Im}(Z) = b$

- Amplitude

  $A = |Z| = \sqrt{(a^2 + b^2)}$

- Phase

  $\Phi = \tan^{-1}(b/a)$

Real and imaginary parts

Math Review - Complex numbers

- Polar Coordinate

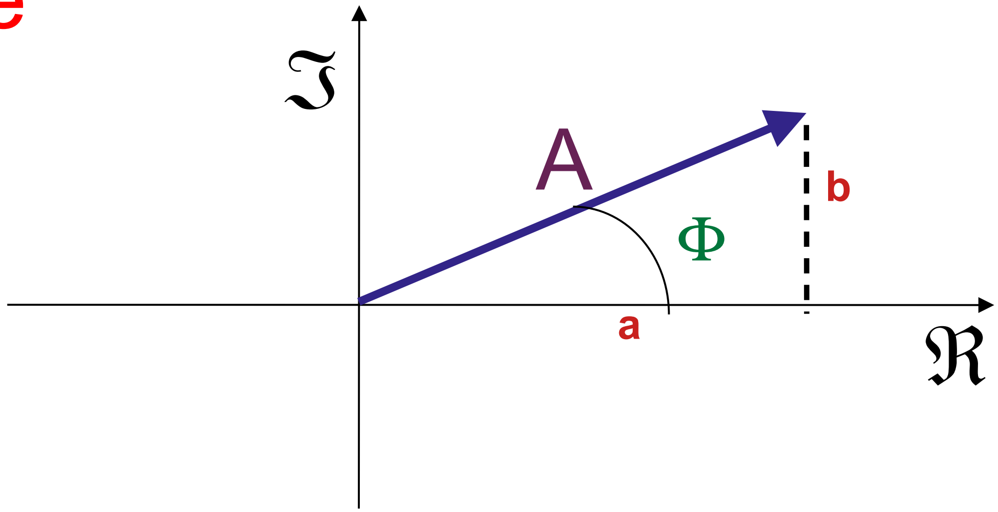
  $Z = a + bi$

- Amplitude

  $A = \sqrt{(a^2 + b^2)}$

- Phase

  $\Phi = \tan^{-1}(b/a)$



Math Review - Complex numbers

- Cosine wave has *three properties*
  - Frequency
  - Amplitude
  - Phase

- Complex number has *two properties*
  - Amplitude
  - Phase

- Use Complex numbers to represent cosine waves at varying frequency
  - Frequency 1: $Z_1 = 5 + 2i$
  - Frequency 2: $Z_2 = -3 + 4i$
  - Frequency 3: $Z_3 = 1.3 - 1.6i$

Math Review - Periodic Functions

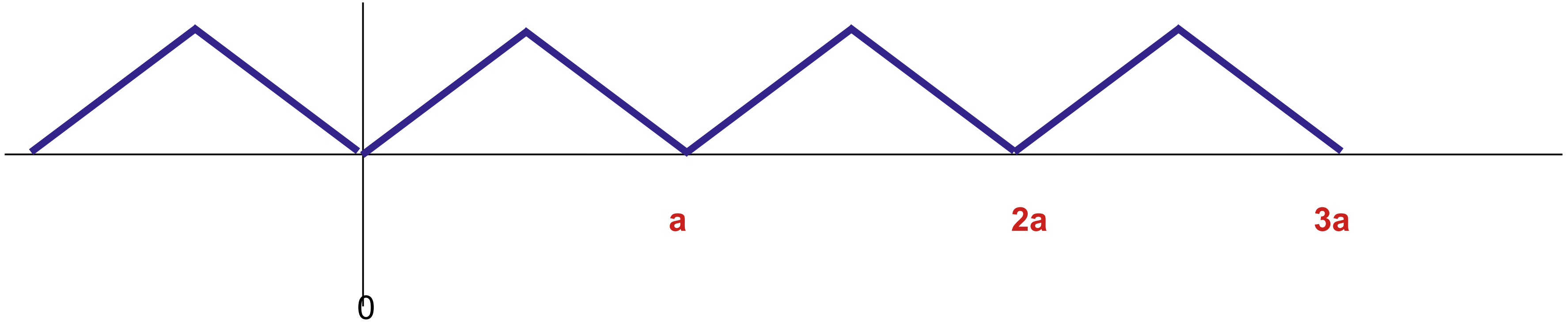
If there is some **a**, for a function **f(x)**, such that

$$\mathbf{f(x) = f(x + na)}$$

then function is periodic *with the period **a***

$$\mathbf{f(x) = f(x + 1a) = f(x + 2a) = f(x + 3a) \ldots\ldots = f(x + na)}$$

(Function Repeats itself with period "a")



Math Review - Attributes of cosine wave

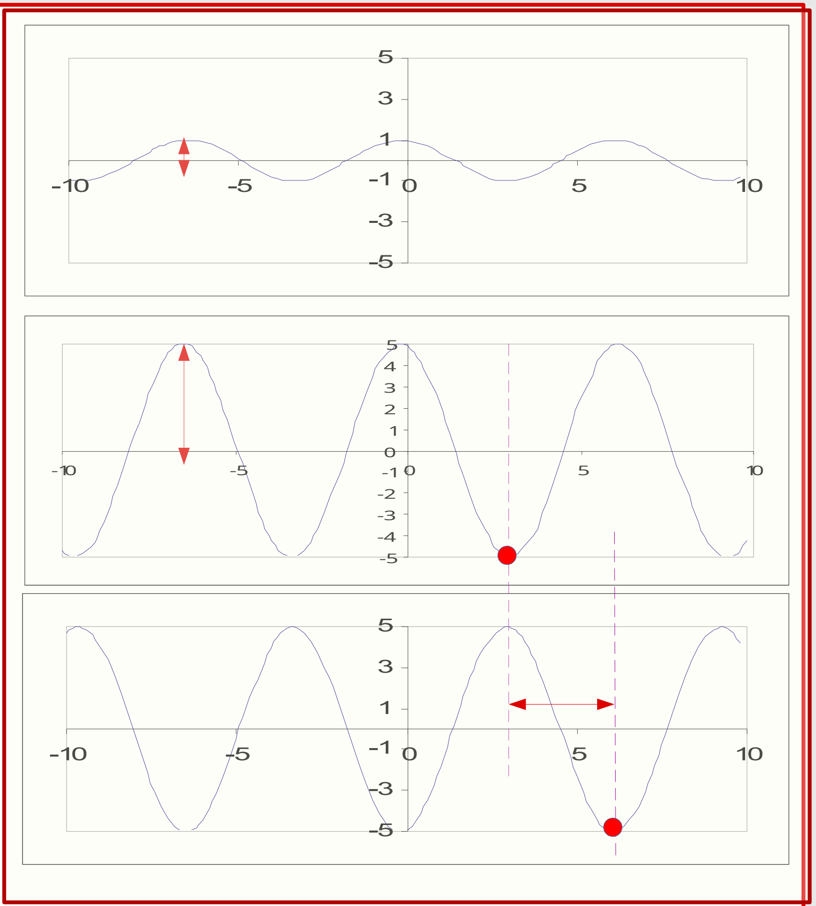
$$f(x) = \cos(x)$$

Amplitude

$$f(x) = \mathbf{5} \cos(x)$$

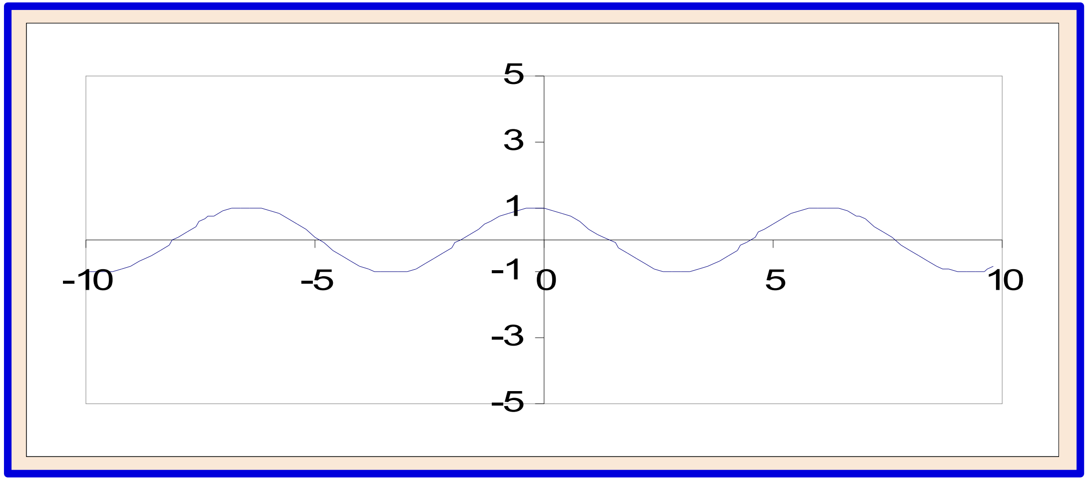
Phase

$$f(x) = 5 \cos(x + \mathbf{3.14})$$



Math Review - Attributes of cosine wave

$f(x) = \cos(x)$



$$f(x) = A \cos(kx + \Phi)$$

Amplitude A, Frequency k, Phase Φ

Fourier Transform Idea

- We want to understand the frequency ω of our signal. So, let's **re-parameterize** the signal by ω instead of *x*:

f(x) → Fourier Transform → ***F(ω)***

- For Each **Freq ω** {from 0 to infinity}, ***F(ω)*** holds the **amplitude *A*** and **phase ϕ** of the corresponding sine $A\sin(\omega x+\phi)$

How can *F* hold both?

Complex number trick! (Complex Number Holds TWO Values) $F(\omega)=R(\omega)+iI(\omega)$

$$A=\pm\sqrt{R(\omega)^2+I(\omega)^2}$$

$$\phi=\tan^{-1}\frac{I(\omega)}{R(\omega)}$$

F(ω) → Inverse Fourier Transform → ***f(x)***

All info are preserved

Low and High Frequency Image Filters

- **Low frequency terms**
  - Close to **origin** in Fourier Space (Origin = 0)
  - Represents smooth areas
- **High frequency terms**
  - Closer to **edge** in Fourier Space (Highest Freq)
  - Necessary to represent edges or high-resolution features

Frequency-based Filters

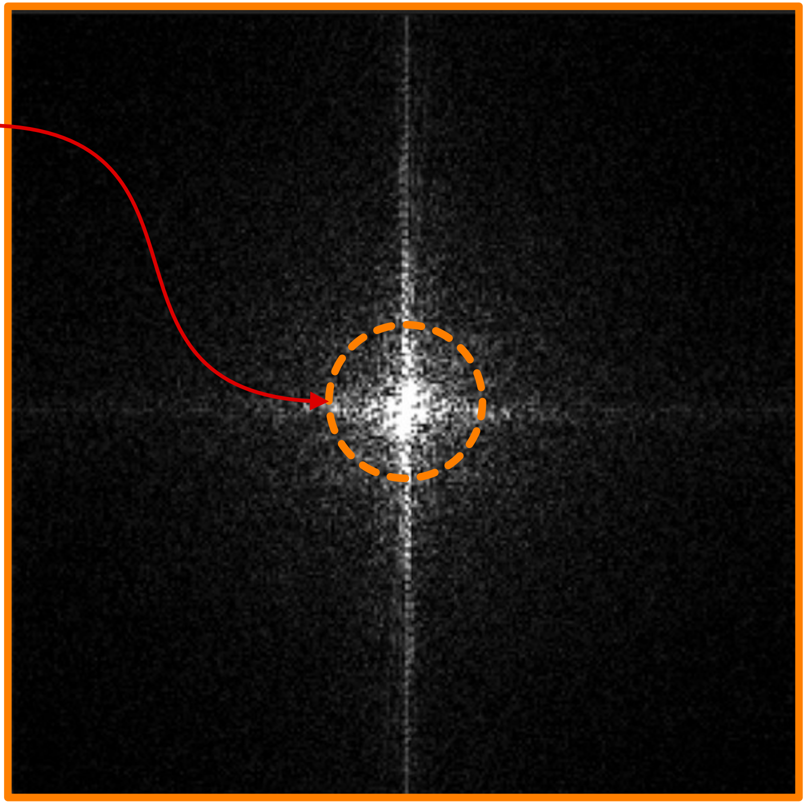
1. **Low-pass Filter** (Blurs – Remove Edges)
   – Restricts data to low-frequency components

2. **High-pass Filter** (Sharpens – Emphasize Edges)
   – Restricts data to high-frequency-components

3. **Band-pass Filter**
   – Restrict data to a band of frequencies

4. **Band-stop Filter**
   – Suppress a certain band of frequencies

Fourier Transform of 2-D Image
from XY Spatial domain to UV Frequency Domain



Representation of Complex Numbers in Exponential form

Euler's Equation $e^{i\,\theta} = \cos\theta + i\sin\theta = \Re(\theta) + \Im(\theta)$

$$
\begin{aligned}
z &= a + ib \\
&= |z| \cdot (\cos\theta + i\sin\theta) \qquad \text{where} \quad |z| = \sqrt{a^2 + b^2} \\
&= |z| \cdot e^{i\theta} \qquad\qquad\qquad\qquad\qquad \theta = \tan^{-1}\left(\frac{b}{a}\right)
\end{aligned}
$$

Complex Number $\mathbf{z} = a + ib = \mathit{Amplitude}\ e^{i\,phase} \Rightarrow |z|\ e^{i\,\theta}$

Introduction to the Fourier Transform and the Frequency Domain

- The two-dimensional Fourier transform **(2-D)** and its inverse
  - **Fourier transform** (discrete case) DTC, Note: **F(u,v) is complex**

$$F(u,v)=\frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)\,e^{-j2\pi(ux/M+vy/N)} \qquad (\textit{for each frequency } u,v)$$

Apply for u=0,1,2,. . . ,M - 1 , for v=0,1,2,..... ,N - 1

  - **Inverse Fourier transform**:

$$f(x,y)=\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)\,e^{j2\pi(ux/M+vy/N)} \qquad (\textit{for each position } x,y)$$

Apply for x=0,1,2,. . . ,M - 1 , for y=0,1,2,..... ,N - 1

- *u*, *v* : the transform or ***frequency*** variables *x*, *y* : the ***spatial*** or image variables

Introduction to the Fourier Transform and the Frequency Domain

- **Fourier is defined in three terms**
- **[1] Spectrum F(u,v)**, **[2] phase angle Φ (u,v)** , and **[3] power spectrum P(u,v)**

$$|F(u,v)| = \left[R^2(u,v) + I^2(u,v) \right]^{\frac{1}{2}} \quad (\text{spectrum})$$

$$\varphi(u,v) = \tan^{-1}\left[\frac{I(u,v)}{R(u,v)} \right] \quad (\text{phase angle})$$

$$P(u,v) = |F(u,v)|^2 = R^2(u,v) + I^2(u,v) \quad (\text{power spectrum})$$

- ***R(u,v)***: the **real** part of *F*(*u*,*v*)
- ***I(u,v)***: the **imaginary** part of *F*(*u*,*v*)

Shift Property of Fourier Transofrm

$$\boxed{f(x,y) \Rightarrow F(u,v)}$$

$$F(u,v)=\frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(ux/M+vy/N)}$$

$$F(u,v)=\frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(ux/M)}e^{-j2\pi(vy/N)}$$

Multiply f(x,y) by $e^{\frac{j2\pi u_0 x}{M}} e^{\frac{j2\pi v_0 y}{N}} \Rightarrow f(x,y)\, e^{\frac{j2\pi u_0 x}{M}} e^{\frac{j2\pi v_0 y}{N}}$

Transform new f(x,y)

$$F(u,v)=\frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{\frac{-j2\pi(u-u_0)x}{M}} e^{\frac{-j2\pi(v-v_0)y}{N}} = F(u-u_0, v-v_0)$$

Conclusion

Multiply $f(x,y)$ by $e^{\frac{j2\pi u_0 x}{M}} e^{\frac{j2\pi v_0 y}{N}}$ (Where u_0, v_0 are any selected values) ***Then Apply FT***

Results is **Same** $F(u,v)$ **[of** $f(x,y)$**]** Shifted by u_0, v_0 $\Rightarrow$ $F(u-u_0, v-v_0)$

Shift Property of Fourier Transofrm

(Shift to Center of Spectrum)

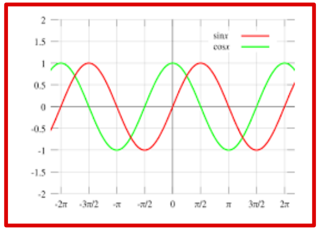
Let $\boxed{u_0 = M/2, v_0 = N/2}$ *Where **M,N** are size of the Image*

$$e^{jx} = \cos(x) + j\sin(x)$$

$$f(x,y)\,e^{\frac{j2\pi u_0 x}{M}}\,e^{\frac{j2\pi v_0 y}{N}} = f(x,y)\,e^{\frac{j2\pi (M/2) x}{M}}\,e^{\frac{j2\pi (N/2) y}{N}} = f(x,y)\,e^{j\pi(x+y)}$$

$$= f(x,y)[\cos\pi(x+y) + j\sin\pi(x+y)] = f(x,y)[(-1)^{(x+y)} + zero]$$

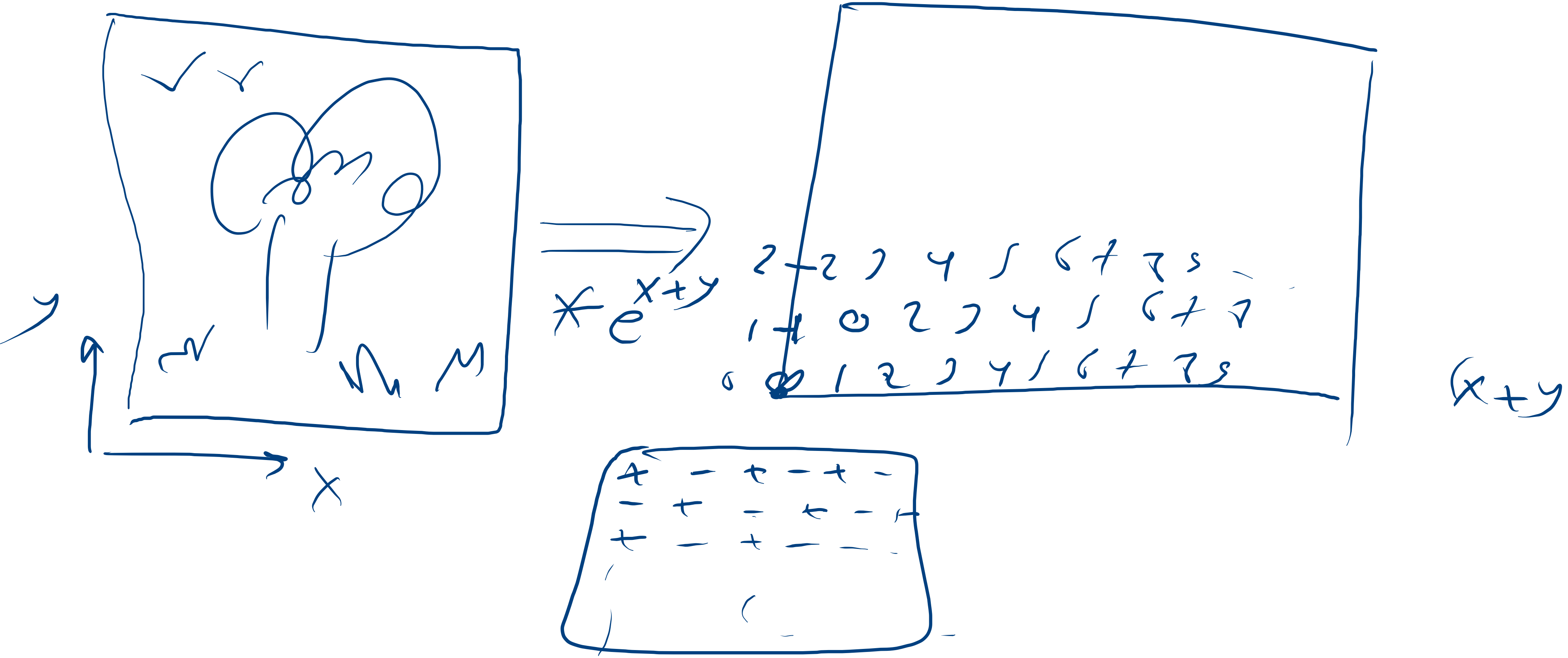
$$= f(x,y)(-1)^{(x+y)} \quad \Rightarrow \quad F(u - M/2, v - M/2)$$



Remember: $\sin(n\pi) = 0 \quad \forall n$

$\cos(n\pi) = 1$ for $n = 0, 2, 4, 6, \ldots$ $\quad = -1$ for $n = 1, 3, 5, 7, \ldots \quad \Rightarrow \quad (-1)^n \quad \forall n$

Multiply Spatial image by $(-1)^{(x+y)}$ Results in shift its spectrum to the center



y
x
$*e^{x+y}$
(x+y

Some Other Properties of Fourier Transofrm

$$F(u,v)=\frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)e^{-j2\pi(ux/M+vy/N)}$$

for $u=0,1,2,\ldots,M-1, v=0,1,2,\ldots,N-1$

$$\Im\left[f(x,y)(-1)^{x+y}\right]=F\left(u-\frac{M}{2},v-\frac{N}{2}\right)$$ (shift $n=M/2, m=N/2$)

$$F(0,0)=\frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)$$ (average: $n=0, m=0$)

$$F(u,v)=F*(-u,-v)$$ (conujgate symmetric)

$$|F(u,v)|=|F(-u,-v)|$$ (symmetric)

Remember: *cos(-u)=cos(u) , sin(-u) = -sin(u)*

Relation between Filtering in the Spatial and Frequency Domain

Convolution theorem:

The discrete convolution of two functions ***f*(*x*,*y*)** and ***h*(*x*,*y*)** of size *MXN* (in Spatial Domain) is defined as:

$$f(x,y)*h(x,y)=\frac{1}{MN}\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}f(m,n)h(x-m,y-n)$$

Filters in Spatial Domain

f(x,y)== Image
h(x,y)== Filter Kernel

Let ***F*(*u*,*v*)** and ***H*(*u*,*v*)** denote the Fourier transforms of ***f*(*x*,*y*)** and ***h*(*x*,*y*)**, then

$$f(x,y)*h(x,y)\Leftrightarrow F(u,v)H(u,v)$$

Convolution in Spatial Domain
==
Multiplication in Frequency Domain

$$f(x,y)h(x,y)\Leftrightarrow F(u,v)*H(u,v)$$

Multiplication in Spatial Domain
==
Convolution in Frequency Domain

Properties of Convolution

- Commutative $$a * b = b * a$$
- Associative $$(a * b) * c = a * (b * c)$$
- Cascade system

$$f \longrightarrow \boxed{h_1} \longrightarrow \boxed{h_2} \longrightarrow g$$

$$= f \longrightarrow \boxed{h_1 * h_2} \longrightarrow g$$

$$= f \longrightarrow \boxed{h_2 * h_1} \longrightarrow g$$

Apply h_2 filter on output of applying h_1 filter on function f

Same Result as if Applying one filter consists of Convolution of h_1 and h_2

Filters Order Does NOT affect final output

Fourier Transform and Convolution

Spatial Domain (*x*)		Frequency Domain (*u*)
$g = f * h$	$\longleftrightarrow$	$G = FH$
$g = fh$	$\longleftrightarrow$	$G = F * H$

we can calculate ***g(x)*** *(Filtered Image)* *by either convolution in spatial domain OR Multiplication in Freq Domain (using* Fourier transform)

f(x,y) conv.

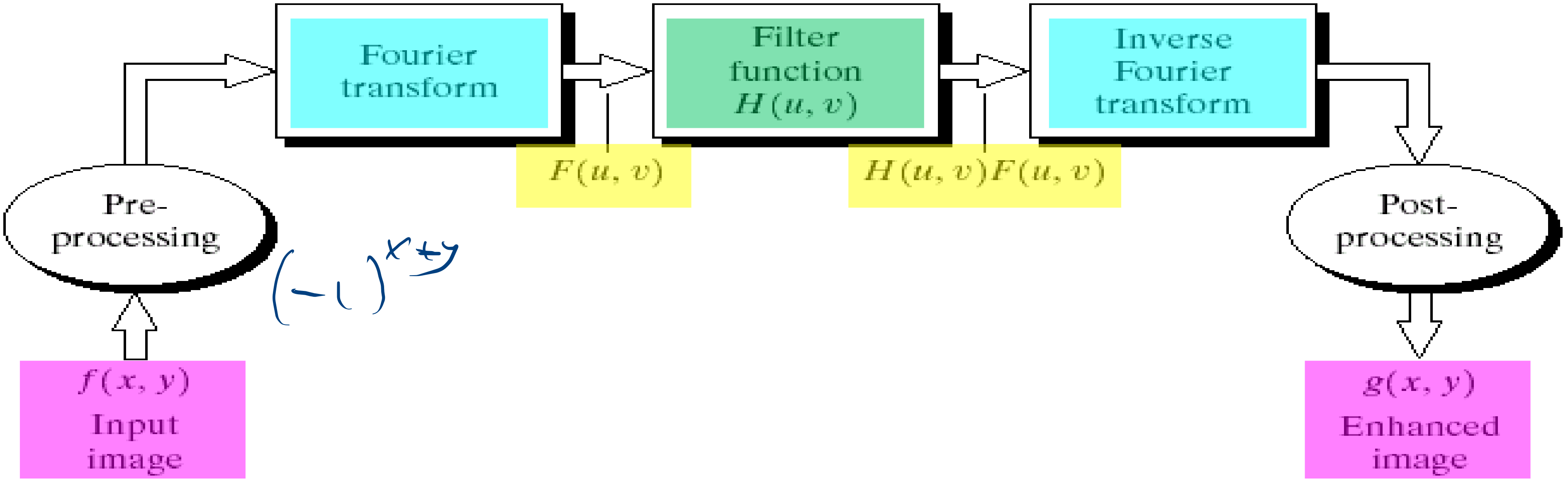
$$g = \underbrace{f}_{f(x,y)} \; \overset{\text{conv.}}{*} \; h$$

Spatial filter

Domain

$$\begin{array}{ccccc} g & = & f & * & h \\ \uparrow \text{IFT} & & \downarrow \text{FT} & & \downarrow \text{FT} \\ G & = & F & \times & H \end{array}$$

Basics of Filtering in the Frequency Domain



Smoothing Frequency-Domain Filters

The basic model for filtering in the frequency domain

$$G(u,v)=H(u,v)F(u,v)$$

where

***F(u,v)*:** the Fourier transform of the **image** to be smoothed

***H(u,v)*:** a **filter** transfer function

Remember: Use Shift Property to move the Origin of Frequency to the center of the image

Smoothing is fundamentally a low pass operation in the frequency domain.

- There are several standard forms of low pass filters (LPF).
  - **Ideal** low pass filter
  - **Butterworth** low pass filter
  - **Gaussian** low pass filter

[1] Ideal Low pass Filters (ILPFs)

The simplest low pass filter is:

A filter that "cuts off" all high-frequency components of the Fourier transform that are **at a distance greater than a specified distance *D_0* from the origin of the transform.**

<u>The transfer function of an ideal low pass filter</u>

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \le D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

One control parameter: **D_0**

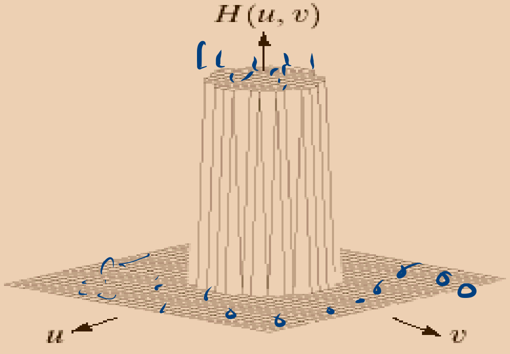
where

***D*(*u*,*v*)** : the distance from point (*u*,*v*) to the **<u>center</u>** of the frequency rectangle

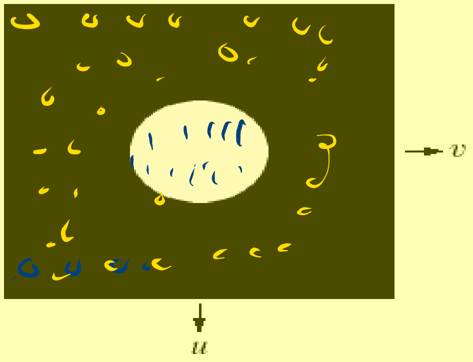
$$D(u,v) = \left[(u - M/2)^2 + (v - N/2)^2\right]^{\frac{1}{2}}$$

D(u,v) is used for **ALL** Filters

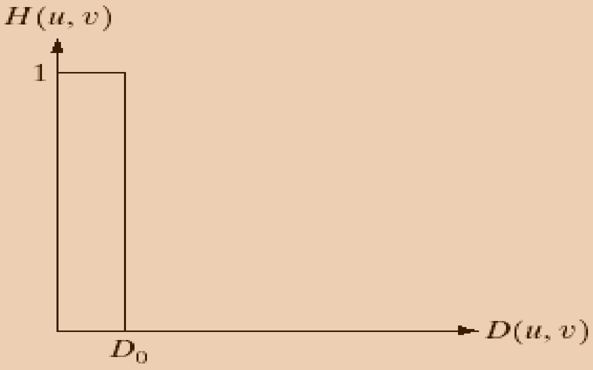
[1] Ideal Lowpass Filters (ILPFs)



Ideal LPF
Transfer Function

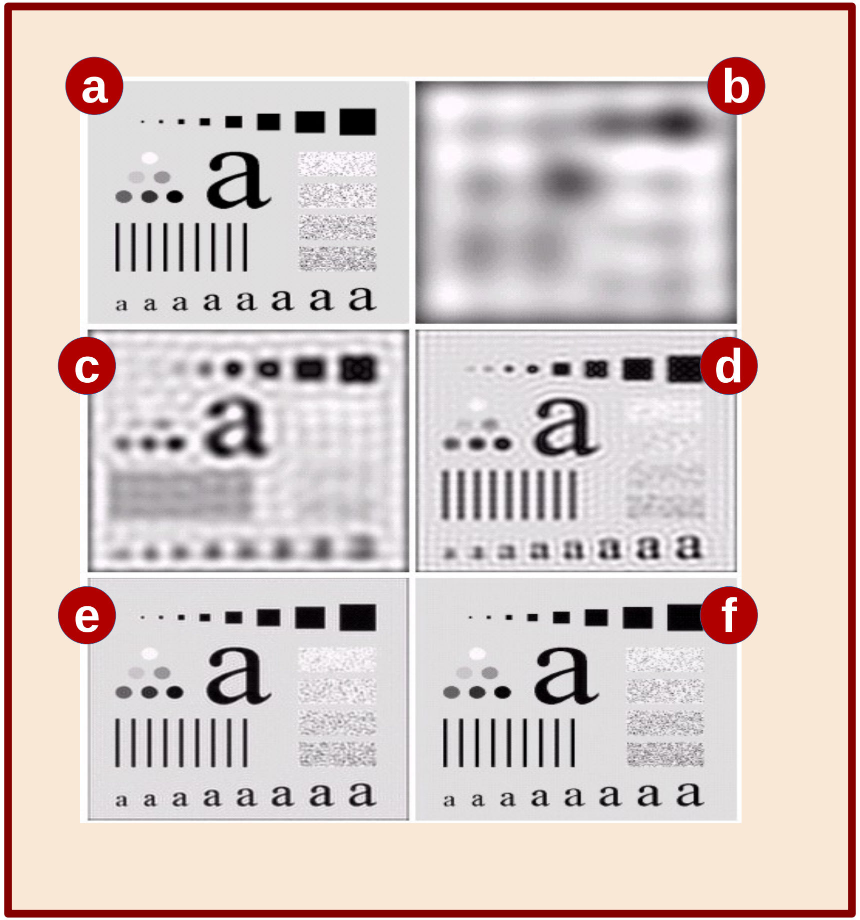


Filter Displayed
as an Image



Filter Radial Cross Section

[1] Ideal Low pass Filters (ILPFs)



(a) Original Image

(b) Low Pass filter at Cutoff Freq. At **radius 5**

(c) Low Pass filter at Cutoff Freq. At **radius 15**

(d) Low Pass filter at Cutoff Freq. At **radius 30**

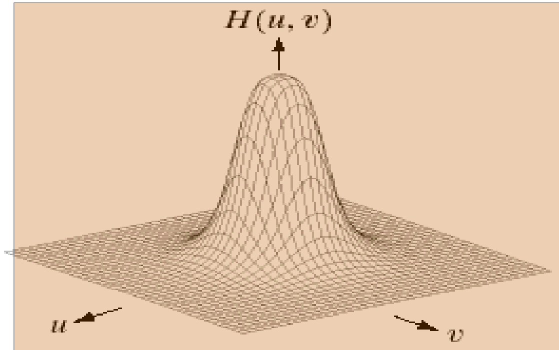
(e) Low Pass filter at Cutoff Freq. At **radius 80**

(f) Low Pass filter at Cutoff Freq. At **radius 230**

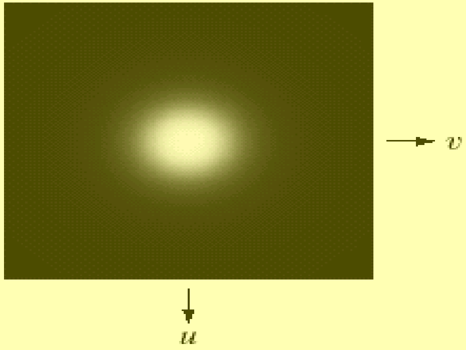
[2] Butterworth Low pass Filters (Order n)

$$H(u,v)=\frac{1}{1+\left[D(u,v)/D_0\right]^{2n}}$$

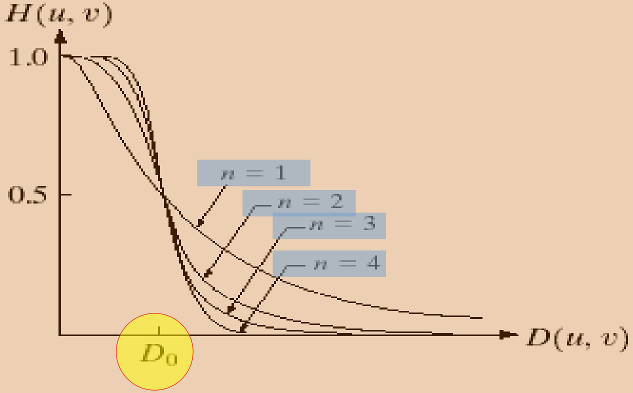
Two control parameters:
D_0 and **n**



Butterworth LPF
Transfer Function

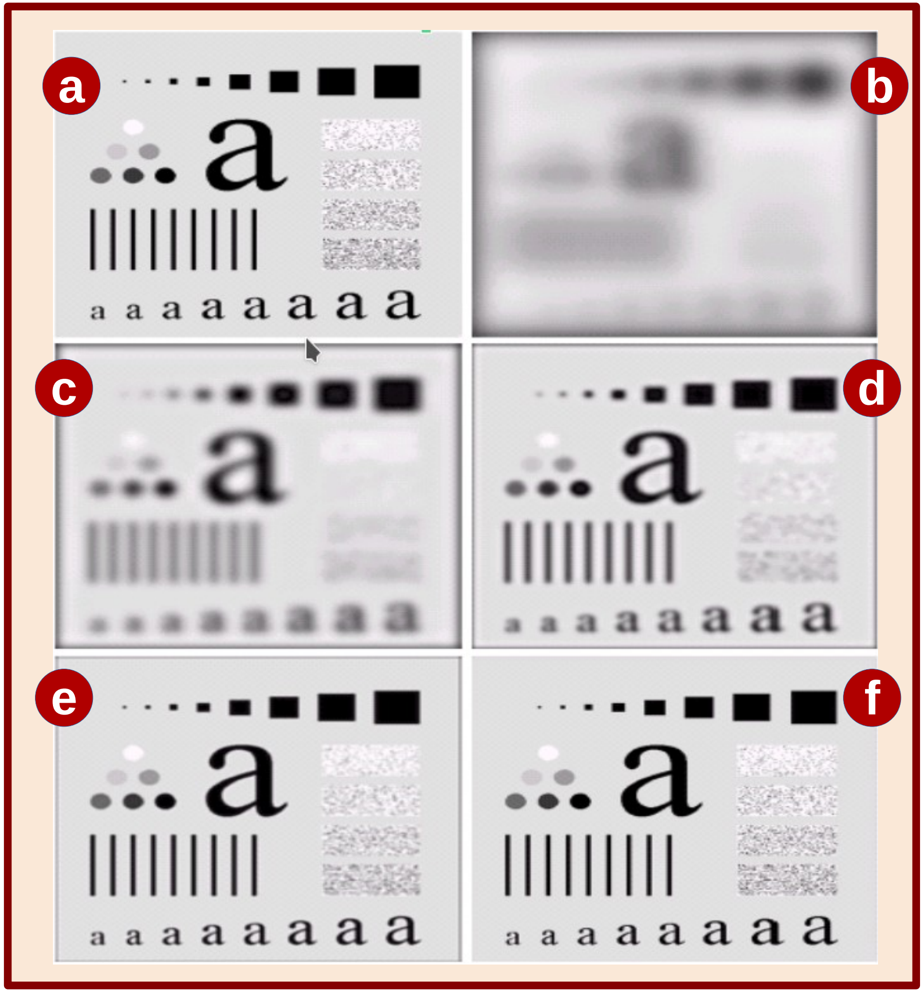


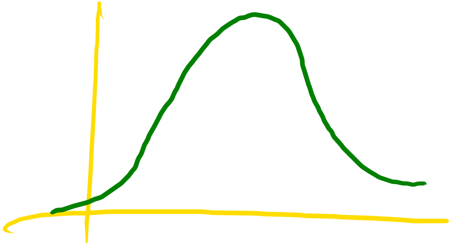
Filter Displayed
as an Image



Filter Radial Cross Section
(Different Orders)

[2] Butterworth Low pass Filters (Order n)





(a) Original Image

(b) Low Pass filter at Cutoff Freq. At **radius 5**

(c) Low Pass filter at Cutoff Freq. At **radius 15**

(d) Low Pass filter at Cutoff Freq. At **radius 30**

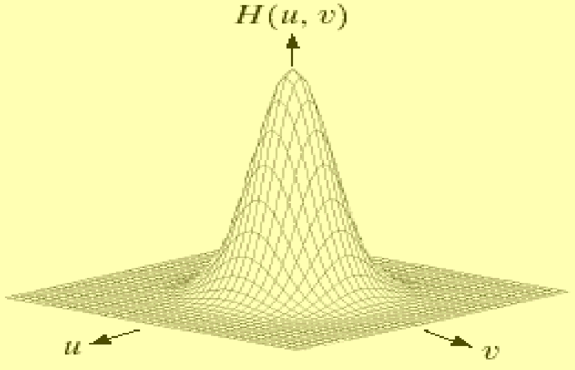
(e) Low Pass filter at Cutoff Freq. At **radius 80**

(f) Low Pass filter at Cutoff Freq. At **radius 230**

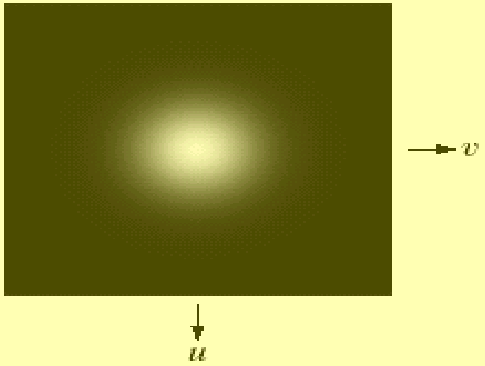
[3] Gaussian Low pass Filters

$$H(u,v) = e^{-D^2(u,v)/2D_0^2}$$

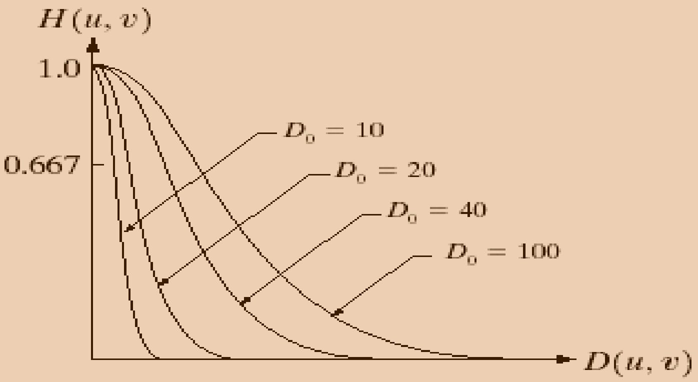
One control parameter: **D_0**



Butterworth LPF
Transfer Function

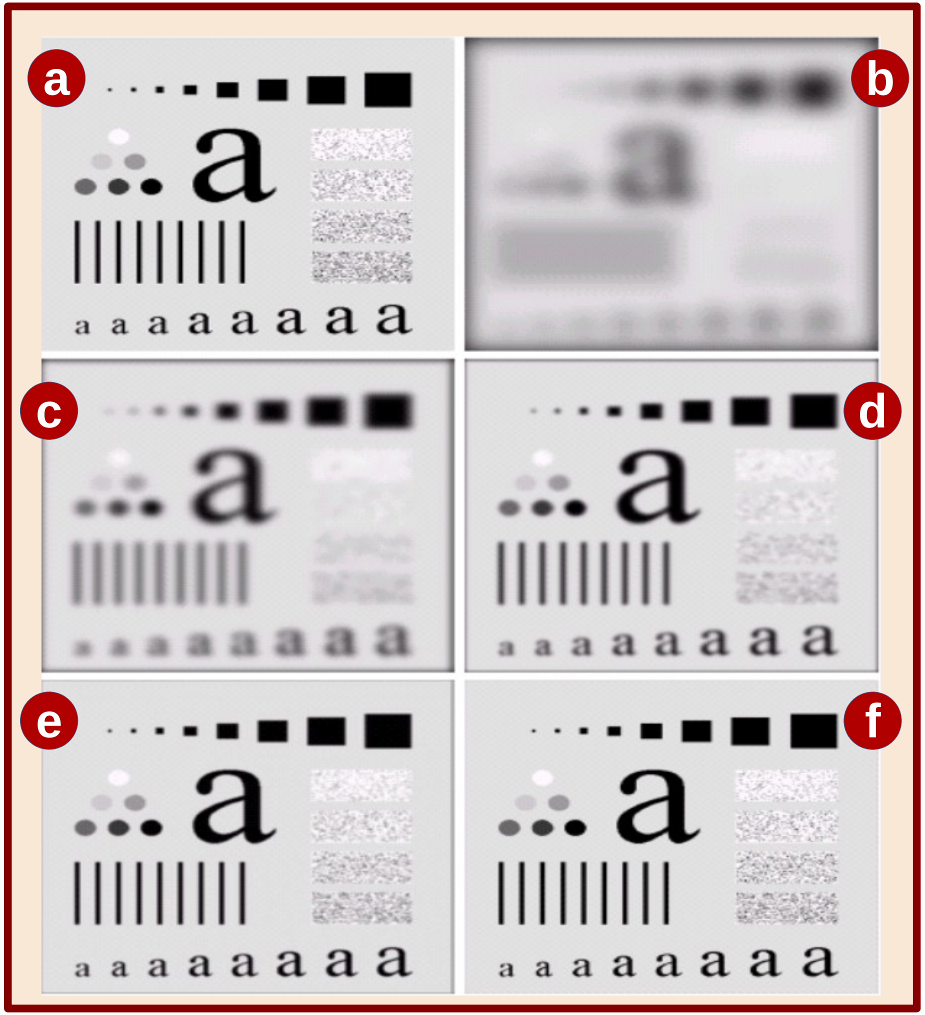


Filter Displayed
as an Image



Filter Radial Cross Section
(Different Orders)

[3] Gaussian Low pass Filters



(a) Original Image

(b) Low Pass filter at Cutoff Freq. At **radius 5**

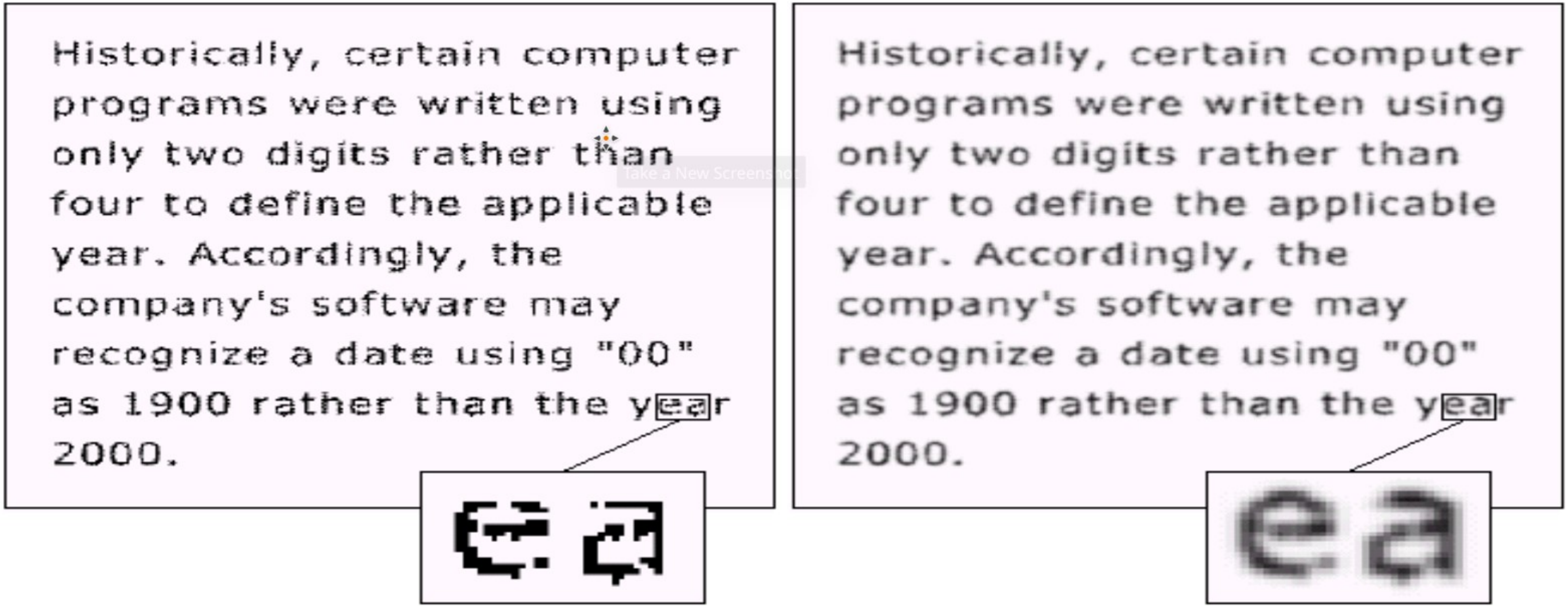
(c) Low Pass filter at Cutoff Freq. At **radius 15**

(d) Low Pass filter at Cutoff Freq. At **radius 30**

(e) Low Pass filter at Cutoff Freq. At **radius 80**

(f) Low Pass filter at Cutoff Freq. At **radius 230**

[3] Gaussian Low pass Filters



Three Different Low pass Filters

[1] Ideal Low pass Filter

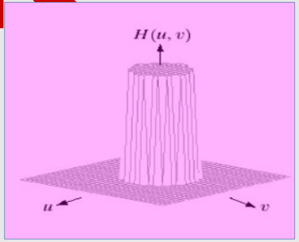
[2] Butterworth Low pass Filter

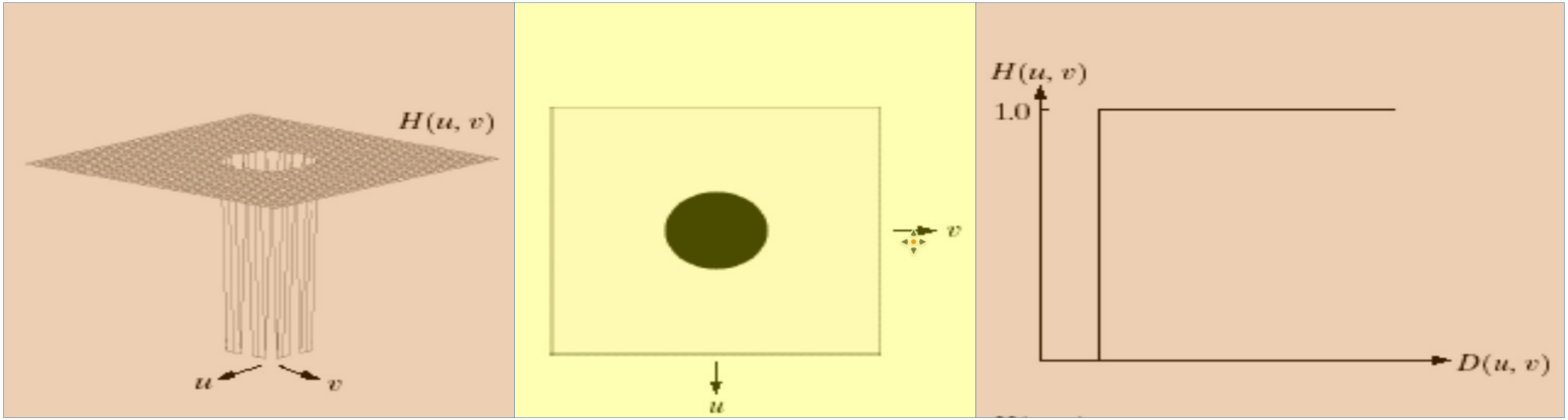
[3] Gaussian Low pass Filter

[1] Ideal High pass Filters

$$H_{hp}(u,v) = 1 - H_{lp}(u,v)$$

$H_{lp}(u,v)$





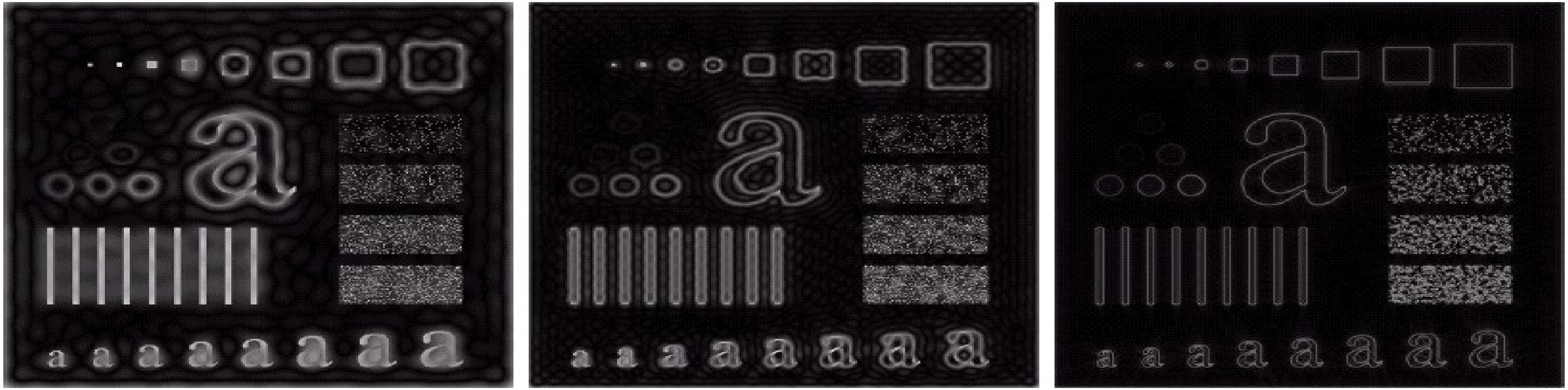
$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \leq D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$$

Remember: Ideal Low Pass Filter (H_{lp})

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \leq D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

[1] Ideal High pass Filters

$$H(u,v)=\begin{cases} 0 & \text{if } D(u,v)\le D_0 \\ 1 & \text{if } D(u,v)> D_0 \end{cases}$$

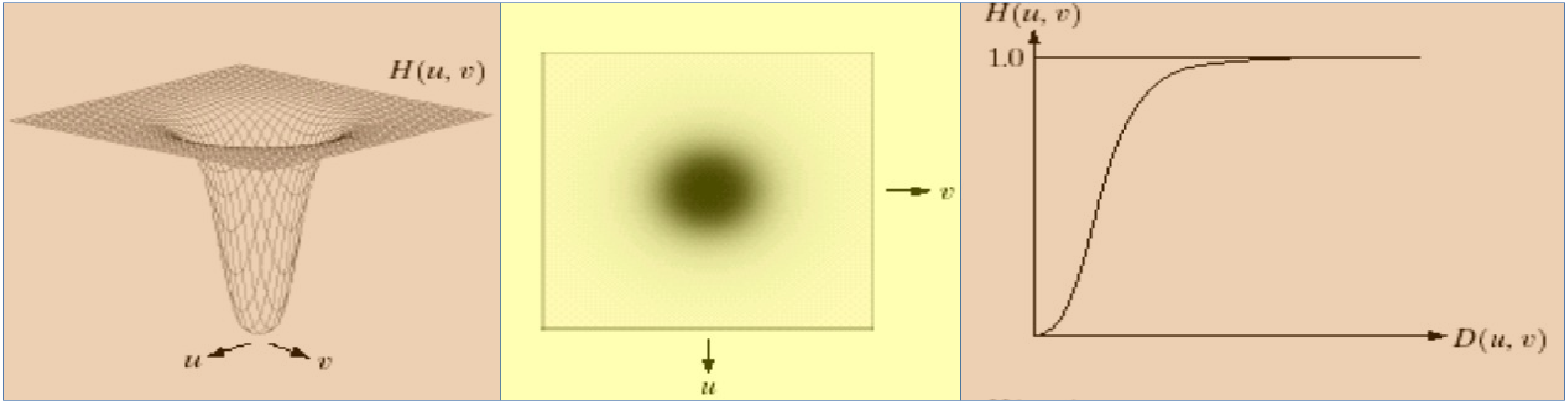


$D_o = 15$ $D_o = 30$ $D_o = 80$

(Ringing effect with D_0 = 15 and 30)

[2] Butterworth High pass Filters

$$H_{hp}(u,v)=1-H_{lp}(u,v)$$



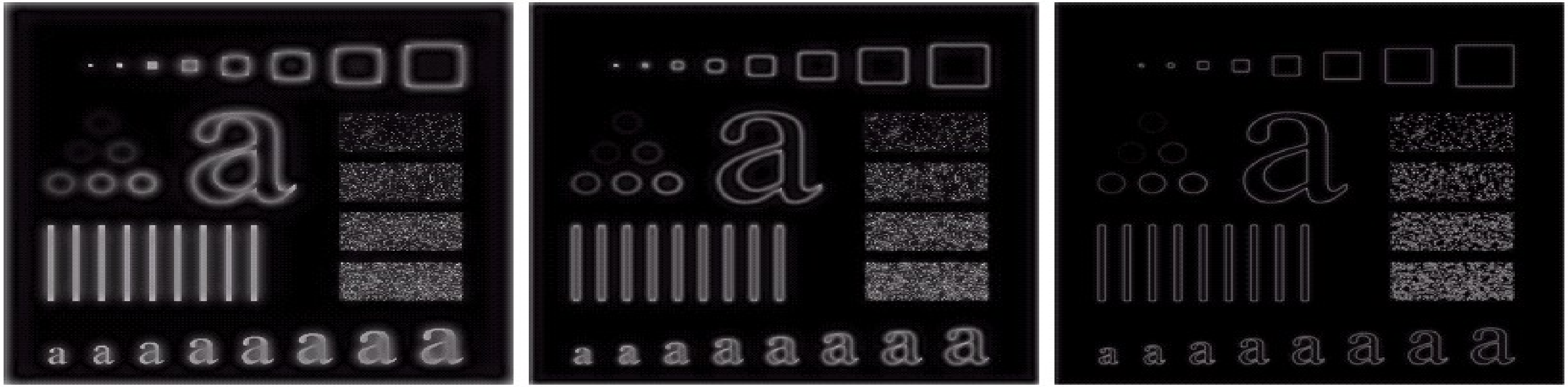
$$H(u,v)=\frac{1}{1+\left[D_0/D(u,v)\right]^{2n}}$$

Remember: Butterworth Low Pass Filter (H_{lp})

$$H(u,v)=\frac{1}{1+\left[D(u,v)/D_0\right]^{2n}}$$

[2] Butterworth High pass Filters

$$H(u,v) = \frac{1}{1 + \left[D_0 / D(u,v)\right]^{2n}}$$

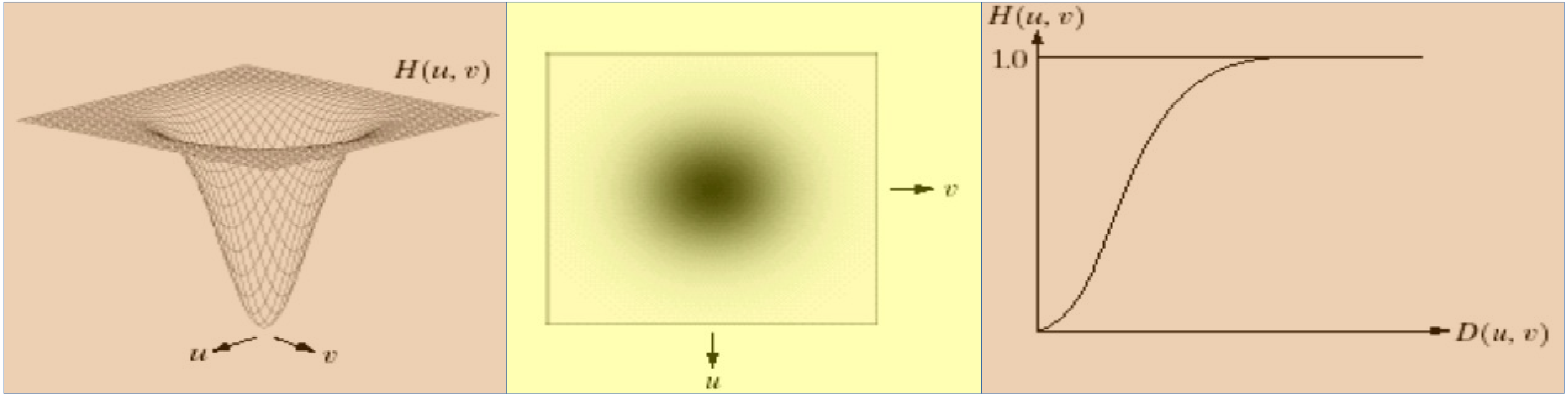


$D_o = 15$ $D_o = 30$ $D_o = 80$

(Much smoother than Results obtained by Ideal HPF)

[3] Gaussian High pass Filters

$$H_{hp}(u,v) = 1 - H_{lp}(u,v)$$



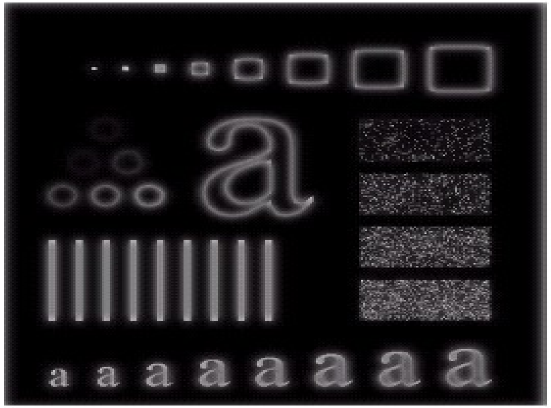
$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$

Remember: Gaussian Low Pass Filter (H_{lp})

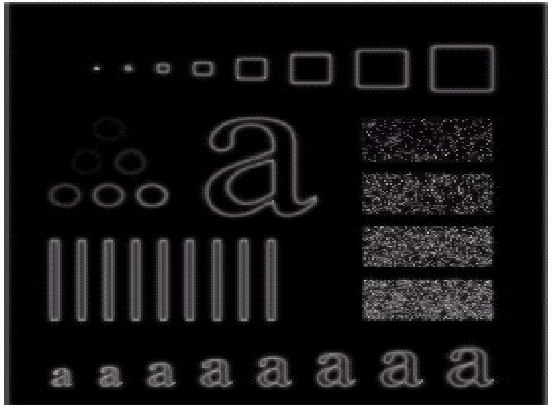
$$H(u,v) = e^{-D^2(u,v)/2D_0^2}$$

Gaussian High pass Filters

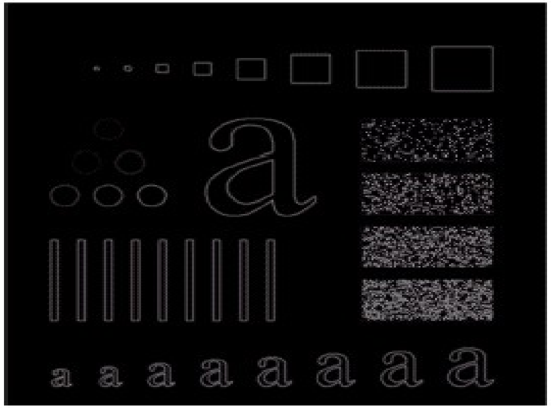
$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$



$D_o = 15$



$D_o = 30$



$D_o = 80$

Sharpening Frequency Domain Filters

Ideal high-pass filter

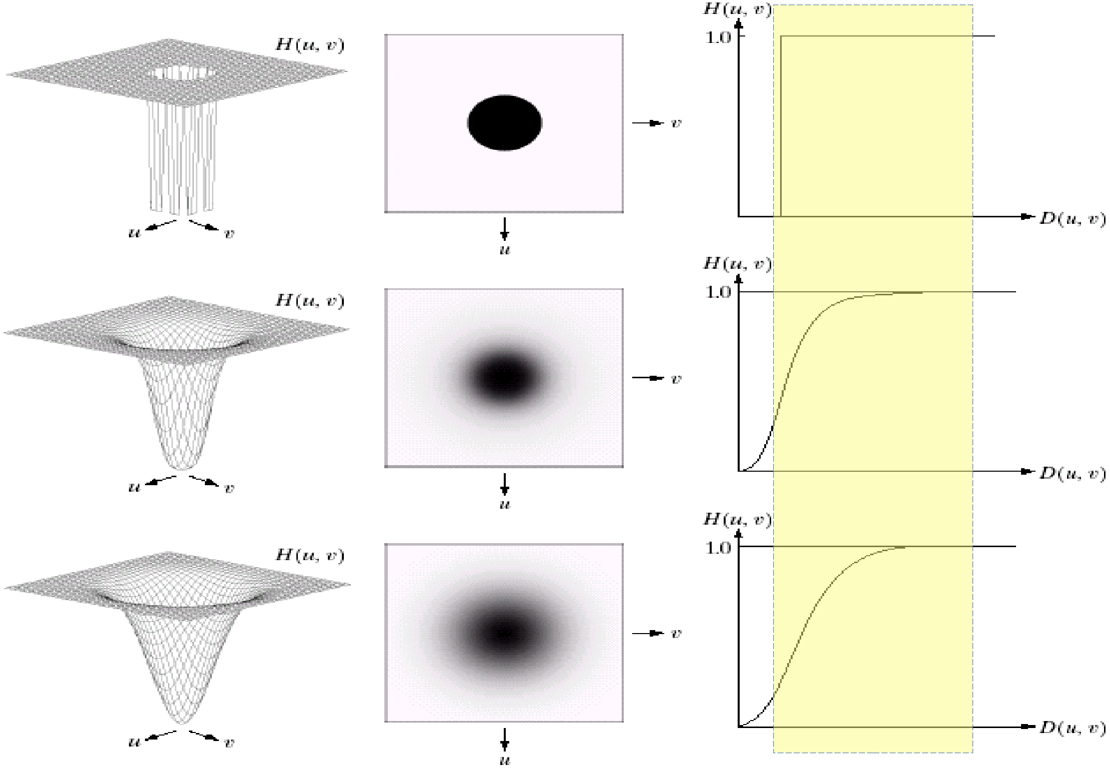
$$H(u,v)=\begin{cases}1 & \text{if } D(u,v)\le D_0\\ 0 & \text{if } D(u,v)>D_0\end{cases}$$

Butterworth high-pass filter

$$H(u,v)=\frac{1}{1+[D_0/D(u,v)]^{2n}}$$

Gaussian high-pass filter

$$H(u,v)=1-e^{-D^2(u,v)/2D_0^2}$$



$$H_{hp}(u,v)=1-H_{lp}(u,v)$$

Frequency-based Filters

1. **Low-pass Filter** (blurs)
   - Restricts data to low-frequency components

2. **High-pass Filter** (sharpens)
   - Restricts data to high-frequency-components

3. **Band-pass Filter**
   - Restrict data to a band of frequencies

4. **Band-stop Filter**
   - Suppress a certain band of frequencies

Explore by yourself